

Data Sourcing Brief and Taxonomic Audit for Career Navigation Systems

The integrity of a stateless, offline-first career decision system relies entirely on the absolute fidelity of its compile-time data. The discovery that the application currently relies on fractional, discontinued, and commercially skewed datasets presents a critical failure point in user trust and developmental routing. The governing principle dictating this data overhaul dictates that simulated or deprecated pathways must be aggressively pruned to prevent catastrophic educational missteps by the student populace.

An overarching analysis of the Indian educational framework reveals a landscape currently undergoing severe tectonic shifts. These shifts are primarily driven by the implementation of the National Education Policy (NEP) 2020, the restructuring of state assessment boards, and the harmonization of vocational frameworks under the National Council for Vocational Education and Training (NCVET). To ensure the application provides genuine, actionable guidance to a demographic ranging from Class 9 to post-graduation, the underlying taxonomy has been completely reconstructed. This reconstruction excises coaching institute products masquerading as official exams, clearly flags stalled governmental schemes, and introduces vital, previously missing corridors such as diploma lateral entry and apprenticeship frameworks.

The Unified Examination Register

The examination register serves as the foundational routing mechanism for the application. A rigorous verification process has identified several critical anomalies in the legacy dataset that actively misguide users.

Most prominently, the National Talent Search Examination (NTSE), historically a cornerstone of Class 10 talent identification, has not been approved beyond the March 2021 cycle by the Ministry of Education¹. The scheme remains indefinitely stalled pending a structural revamp intended to enhance visibility among rural students¹. The NCERT explicitly confirms that the scheme in its present form has not been approved for further implementation¹. Continuing to present the NTSE as an active pathway represents a severe systemic hazard; it has thus been flagged with a suspended status.

Similarly, entry into the Rashtriya Indian Military College (RIMC) is strictly limited to Class 8 students within a narrow age band, rendering it an invalid suggestion for older cohorts³. The All India Sainik Schools Entrance Examination (AISSEE) operates on strict age parameters for Class 6 (10-12 years) and Class 9 (13-15 years) entry, making it inaccessible beyond those stages⁴.

Furthermore, the register now incorporates lateral entry examinations, which represent the primary upward mobility vector for diploma holders seeking engineering degrees. The absence of state-level lateral entry tests (such as OJEE Lateral Entry, AP POLYCET, and TS POLYCET) previously severed the digital progression of vocational students⁶.

The data presented in the tables below fulfills Deliverable 1 and is structured as the

CSV-equivalent data source for direct compilation into Dart map keys.

Secondary School and Military Entry Examinations

id	ful IN a m e	co nd uc tin g B od y	st at us	le ve l	ap pli ca bl eS ta ge s	eli gi bil ity	ca te go ry R el ax ati on s	re gi st ra tio nF ee	ty pi ca lW in do w	att e m pt sP er Ye ar	of fic ial Ur l	so ur ce Ur l	la st Ve rif ie d At	D es cri pti on (F or 15 yo)
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Polytechnic, Lateral Entry, and Vocational Pathways

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p_ po lyc et	ad es h Po lyt ec hn ic C o m m on En tra nc e Te st	T A P			0	pa ss ed /a pp ea rin g; mi ni m u m 35 % m ar ks .	fic ati on : tru e	fic ati on : tru e	ay		.ni c.i n		03	nc e ex a m for 3- ye ar en gi ne eri ng di pl o m as in An dh ra Pr ad es h rig ht aft er Cl as s 10 .
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po lyc et	St at e Po lyt ec hn ic C o m m o n En tra nc e Te st	T S				ss ed /a pp ea rin g; m at he m ati cs , sci en ce fo cu s.	ati on : tru e	ati on : tru e			.te la ng an a. go v.i n		03	en ter a po lyt ec hn ic co lle ge for an en gi ne eri ng di pl o m a in Te la ng an a.
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ts	pr en tic es hi p Tr ai ni ng Sc he m e				un de rgr ad ua te	Di pl o m a ho ld er s in en gi ne eri ng /te ch no lo gy .	ati on : tru e	en d pr ov id ed)	d		on .g ov .in		03	ng sc he m e for st ud en ts w ho ha ve co m pl et ed th eir di pl o m a or de gr ee an d ne ed w or k ex pe rie nc
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Undergraduate, Graduate, and Professional Examinations

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						s).								s lik e NI Ts , and the fir st st ep for IIT s.
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														ge pr of es so r or ge ta go ve rn m en t re se ar ch gr an t.
ex a m _c sir _n et	Jo int C SI R- U G C N E T	N TA / C SI R	ac tiv e	na tio na l	po st gr ad ua te	M. Sc or eq ui va le nt; mi ni m u m 55 % m ar ks	ne ed sV eri fic ati on : tru e	ne ed sV eri fic ati on : tru e	Ju n, D ec	2	csi rn et. nt a. ac .in	²¹	20 26 -0 8- 03	Si mi lar to U G C N E T, bu t sp ec ific all y for st

						.								ud en ts w ho st ud ie d Sc ie nc e su bj ec ts lik e Ph ysi cs , C he mi str y, an d Lif e Sc ie nc es .
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ex a m _c ds	C o m b i n e d D e f e n c e S e r v i c e s E x a m i n a t i o n	U P S C	ac t i v e	na t i o n a l	gr a d u a t e	D e g r e e f r o m r e c o g n i z e d u n i v e r s i t y ; s p e c i f i c a g e s p e r a c a d e m y.	ne e d s V e r i f i c a t i o n : t r u e	ne e d s V e r i f i c a t i o n : t r u e	Ap r, S e p	2	up sc .g o v .in	²⁵	20 26 -0 8- 03	An e x a m t a k e n a f t e r c o l l e g e t o b e c o m e a n o f f i c e r i n t h e A r m y, N a v y, o r A i r F o r c e.
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_n da _n a	al D ef en ce Ac ad e m y & N av al Ac ad e m y Ex a m	C		I	2, dr op pe r	12 ap pe ari ng /p as se d; un m arr ie d; str ict ph ysi ca l st an da rd s.	eri fic ati on : tru e	eri fic ati on : tru e	p		ov .in		8- 03	a m to joi n th e mi lit ar y str ai gh t ou t of hi gh sc ho ol, le ad in g to a ca re er as a co m mi ssi on ed off ic
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Relational Two-Tier Interest Taxonomy

The legacy architecture featured a flat list of thirty unstructured strings. This forced users into severe cognitive overload, artificially mixing broad industries with highly specific, jargon-heavy job titles. For a 15-year-old student, the distinction between "Data & AI" and "Computers & IT" is meaningless without context; it is a distinction understood only by industry insiders.

To rectify this, the taxonomy has been restructured into a two-tier relational model. The primary clusters represent macro-fields immediately recognizable to a Class 9 demographic. Beneath these clusters, specific interests map dynamically to the streams, subjects, and examination pathways identified in Deliverable 1. This structure ensures that vocational and non-elite trajectories—such as skilled trades, hospitality, and local government work—are granted parity with elite academic routes. This directly addresses the workforce imbalances highlighted in macroeconomic policy reports, which note that 88% of the Indian workforce operates in roles misaligned with higher educational degrees, stressing the urgent need for visible technical and vocational education and training (TVET) pathways²⁷.

Macro-Cluster 1: Engineering, Making, and Skilled Trades

This cluster aggregates advanced engineering degrees and foundational industrial trades, establishing a seamless continuum of mechanical and technical interests without stigmatizing vocational routes.

id	interest Name	streams	subjects	courses	exams	careerExamples	dayToDay
int_mech_auto	Machines & Vehicles	PCM, Vocational	Mathematics, Physics	B.Tech Mech, Diploma Auto, ITI Fitter	exam_jee_main, exam_ap_polycet	Mechanical Engineer, Auto Mechanic	Designing, repairing, and testing engines and large physical machines.
int_electrical	Electrical & Wiring	PCM, Vocational	Physics, Mathem	B.Tech EE, ITI Electrici	exam_jee_expo, exam_o	Electrician, Power	Installing wiring,

			atics	an	jee_le	Grid Engineer	repairing electronics, and managing power systems.
int_construction	Building & Infrastructure	PCM, Vocational	Physics , Mathematics	B.Tech Civil, Diploma Civil	exam_jee_main, exam_tss_polycet	Civil Engineer, Construction Supervisor	Planning structures, surveying land, and managing building sites.

Macro-Cluster 2: Healthcare, Biology, and Agriculture

This cluster merges traditional medical pathways with allied health, veterinary, and modern agricultural sciences, reflecting a diverse ecosystem of biological careers.

id	interest Name	streams	subjects	courses	exams	careerExamples	dayToDay
int_medicine	Human Health & Medicine	PCB	Biology, Chemistry, Physics	MBBS, BDS, B.Sc Nursing	exam_neet_ug	Doctor, Nurse, Paramedic	Diagnosing illness, treating patients , and managing health crises.

int_agri_farming	Farming & Agriculture	PCB, PCMB, Vocational	Biology, Agriculture	B.Sc Agriculture, ITI Agro	exam_j expo	Agricultural Scientist, Farm Manager	Improving crop yields, managing soil health, and running farm operations.
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Macro-Cluster 3: Computing, Data, and Information Technology

Abstracting convoluted tech jargon into an accessible format, this cluster focuses on digital creation, software logic, and hardware networking.

id	interest Name	streams	subjects	courses	exams	career Examples	dayToDay
int_software	Coding & Software	PCM	Computer Science , Math	B.Tech CS, BCA, MCA	exam_j ee_main, exam_u gc_net	Software Developer, App Designer	Writing code, building software applications, and solving logic puzzles.
int_hardware_net	IT & Networking	PCM, Vocational	Computer Science , Physics	Diploma IT, ITI COPA	exam_a p_polycet	Network Admin, IT Support	Fixing computers, setting up internet networks, and

							managing servers.
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Macro-Cluster 4: Governance, Defense, and Administration

Capturing the high aspiration for government service, uniformed defense, and administrative stability within the Indian demographic.

id	interest Name	streams	subjects	courses	exams	careerExamples	dayToDay
int_defense	Military & Uniformed Forces	Any	Mathematics, General Studies	NDA, CDS, B.A/B.Sc	exam_n da_na, exam_c ds	Army Officer, Air Force Pilot	Leading troops, maintaining physical fitness, and protecting national security.
int_civil_serv	Government & Civil Service	Arts, Commerce, Science	Political Science , History	B.A, B.Com, B.Sc	exam_u gc_net, UPSC (Placeholder)	IAS Officer, Local Govt Clerk	Managing public policy, processing citizen services , and administrative work.

Macro-Cluster 5: Commerce, Finance, and Retail

Focusing on the mechanics of money, trade, and enterprise.

id	interest Name	streams	subjects	courses	exams	careerExample s	dayToDay
int_banking_fin	Banking & Accounts	Commerce	Accountancy, Economics	B.Com, CA, BBA	exam_sbi_po	Bank Manager, Accountant	Managing financial ledgers, auditing accounts, and advising on investments.
int_sales_retail	Sales & Retail Management	Any, Vocational	Business Studies	BBA, Diploma Retail	scheme_naps	Store Manager, Sales Executive	Managing shops, interacting with customers, and tracking inventory.

State Boards, Subject Matrices, and NEP Reforms

The structural mapping of regional educational boards requires high precision, particularly where governance is split between secondary and higher secondary levels. Furthermore, the transition toward the National Education Policy (NEP) 2020 framework is altering examination frequencies and subject rigidities.

The Changing Architecture of State Boards

A historic error in national datasets is the conflation of state board authorities. The state of Karnataka exemplifies this complexity. Historically, Class 10 (SSLC) was governed by the Karnataka Secondary Education Examination Board (KSEEB), while Class 11 and 12 (PUC) were governed by the Department of Pre-University Education (DPUE). In 2022, these entities were officially merged to form the Karnataka School Examination and Assessment Board (KSEAB)²⁹. The KSEAB now administers the entire secondary and higher secondary framework

and has pioneered the implementation of 'three annual examinations' (Exam 1, Exam 2, Exam 3) to replace traditional supplementary exams³⁰.

Similarly, Andhra Pradesh maintains a strict bifurcation between the Board of Secondary Education Andhra Pradesh (BSEAP) for Class 10 and the Board of Intermediate Education Andhra Pradesh (BIEAP) for Class 11 and 12³¹. West Bengal mirrors this with the West Bengal Board of Secondary Education (WBBSE) and the West Bengal Council of Higher Secondary Education (WBCHSE)³³.

code	name	state	levels	officialUrl	sourceUrl	lastVerifiedAt
brd_kar_kseab	Karnataka School Examination and Assessment Board	Karnataka	class10, class11, class12	kseab.karnataka.gov.in	²⁹	2026-08-03
brd_ap_bseap	Board of Secondary Education Andhra Pradesh	Andhra Pradesh	class10	bse.ap.gov.in	³¹	2026-08-03
brd_ap_bieap	Board of Intermediate Education Andhra Pradesh	Andhra Pradesh	class11, class12	bie.ap.gov.in	³²	2026-08-03
brd_wb_wbbse	West Bengal Board of Secondary Education	West Bengal	class10	wbbse.wb.gov.in	³³	2026-08-03

brd_wb_wbchse	West Bengal Council of Higher Secondary Education	West Bengal	class11, class12	wbchse.wb.gov.in	³³	2026-08-03
brd_bih_bbose	Bihar Board of Open Schooling and Examination	Bihar	class10, class12	bbose.org	³⁵	2026-08-03
brd_nat_cbse	Central Board of Secondary Education	National	class10, class11, class12	cbse.gov.in	³⁶	2026-08-03

NEP Dual Board Exams Implementation

Following the Ministry of Education's directive aligned with the NEP 2020, the Central Board of Secondary Education (CBSE) will institute a dual-board examination system beginning in the 2025-26 academic session³⁶. The first examination phase will occur between mid-February and March (specifically slated to begin the first Tuesday after February, spanning 34 days across 84 subjects), acting as the primary evaluation³⁸. A secondary phase will be conducted in May³⁶. Crucially, this framework is not a semester system; both examinations cover the full curriculum, allowing students to retain the best score across the two attempts, thereby mitigating high-stakes pressure³⁶. There will be no separate supplementary exams; students failing 1 to 5 subjects in the first exam are placed in an 'Improvement Category' and must take the second exam³⁸. The implementation of this schedule requires the application to shift from a singular timeline assumption for CBSE students to a dual-window preparation strategy.

Rigid Subject Matrices: The West Bengal Anomaly

Relying on implicit inference to map streams to subjects produces critical routing errors for state boards utilizing strict combinatorial sets. The West Bengal Council of Higher Secondary Education (WBCHSE) operates on a highly specific semesterized elective matrix³⁴. Students are

required to select three compulsory electives and one optional elective restricted strictly within one of three defined sets³⁴.

Under Set I (primarily Science and Technical routes), students must choose between explicitly paired alternatives, such as deciding exclusively between Physics or Nutrition, and between Chemistry or Geography³⁴. Recently introduced technological fields, such as Cyber Security (CBST) and Artificial Intelligence & Data Science (AIDS), are also strictly governed within these sets⁴¹. Set II is explicitly curated for Commerce and Business administration, heavily featuring Accountancy, Business Studies, and highly specific alternatives like Commercial Law versus Statistics³⁴. Set III encompasses the Humanities and Social Sciences⁴¹. A digital architecture must statically enforce these rules; failure to do so will guide students toward subject combinations that the state government actively prohibits, invalidating their registration trajectory.

board	stream	offeredCombinations	mathOptionality	notes
brd_wb_wbchse	Science (Set I)	Physics OR Nutrition; Chemistry OR Geography; Math OR Agriculture; Bio; CS OR Modern App OR EVS.	Yes (Can choose Agriculture instead)	Strict pairwise exclusionary combinations ³⁴ .
brd_wb_wbchse	Commerce (Set II)	Accountancy; Business Studies; Commercial Law OR Stats; Costing & Taxation.	Yes (Through Business Math & Stats)	Cannot mix with core Set I sciences ³⁴ .
brd_wb_wbchse	Humanities (Set III)	Pol Science OR Bio; Education OR Nutrition; History OR Psych OR	Yes (Through Basic Math for Social Sciences)	Massive flexibility but strict OR conditions ⁴¹ .

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Vocational Integration: ITI Trades and Diploma Branches

Vocational education in India has historically suffered from informational asymmetry, leading to a societal undervaluation of critical trades. The National Skills Qualification Framework (NSQF), managed by the National Council for Vocational Education and Training (NCVT), acts as the primary standardizing matrix, grading competencies from Level 1 (lowest complexity) to Level 8 (highest complexity)⁴².

The Directorate General of Training (DGT) manages the Industrial Training Institutes (ITIs), which predominantly operate at NSQF Levels 3, 4, and 5⁴⁵. Integrating this data resolves the systemic blind spot surrounding the transition from vocational training to formal higher education, a progression heavily advocated for by NITI Aayog policy reports targeting the optimization of India's demographic dividend²⁷.

Industrial Training Institute (ITI) Trade Register

The application previously offered only 28 fictionalized trades. The actual DGT framework operates over 137 NSQF aligned trades across approximately 15,000 ITIs⁴⁶.

tradeId	tradeName	duration	entryQualification	nsqfLevel	affiliation	lateralProgression
iti_electrician	Electrician	2 Years	Class 10 with Science/Math	Level 4	NCVT / SCVT	Diploma Electrical Engg. ⁴⁷
iti_fitter	Fitter	2 Years	Class 10 with Science/Math	Level 4	NCVT / SCVT	Diploma Mechanical Engg. ⁴⁷
iti_copa	Computer Operator & Prog. Assistant (COPA)	1 Year	Class 10	Level 3	NCVT / SCVT	Diploma Computer Science ⁴⁷

iti_solar_tech	Solar Technician (Electrical)	1 Year	Class 10 with Science/ Math	Level 3	NCVT	Associated Apprenticeships ⁴⁵
iti_welder	Welder	1 Year	Class 8	Level 3	NCVT / SCVT	Specialized Apprenticeships ⁴⁷

AICTE Diploma Branch Register

Diploma lateral entry routes represent a massive demographic flow that is currently unserved by the app. These AICTE-recognized branches lead directly into Bachelor of Technology degrees via state-level examinations.

branchId	branchName	durationStandard	entryRoute	lateralRoute	leadsToDegree
dip_mech	Diploma in Mechanical Engineering	3 Years	Class 10	Class 12 / ITI Fitter (2 Yrs)	B.Tech Mechanical ⁶
dip_civil	Diploma in Civil Engineering	3 Years	Class 10	Class 12 / ITI Draftsman (2 Yrs)	B.Tech Civil ⁶
dip_ee	Diploma in Electrical Engineering	3 Years	Class 10	Class 12 / ITI Electrician (2 Yrs)	B.Tech Electrical ⁶
dip_auto	Diploma in Automobile Engineering	3 Years	Class 10	Class 12 / ITI Motor Mech (2 Yrs)	B.Tech Automobile ⁶

Financial Assistance and Scholarship Telemetry

The data strategy surrounding financial aid requires transitioning away from early friction. Collecting highly sensitive demographic and financial data during initial onboarding drastically inflates cognitive load and attrition rates. Instead, the application must store the metadata of available schemes and surface relevant portals just-in-time when a student explores heavily monetized pathways.

A prominent adjustment in the federal scholarship ecosystem involves the PM Young Achievers Scholarship Award Scheme for Vibrant India (PM YASASVI). The Ministry of Social Justice and Empowerment previously utilized the NTA to conduct the Yasasvi Entrance Test (YET).

However, this examination has been officially discontinued and cancelled⁵¹. The selection criteria have transitioned entirely to a meritocratic evaluation of grades secured in the 8th and 10th-grade final examinations for progression into the 9th and 11th-grade scholarship tiers respectively⁵¹. The scholarship targets students from OBC, EBC, and DNT categories whose parental income does not exceed Rs. 2.5 Lakhs annually, providing substantial aid ranging from Rs. 75,000 to Rs. 1,50,000⁵².

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schol_ pm_ya sasvi	PM YASA SVI Schola rship	MSJ& E	OBC/ EBC/D NT; Class 9 or 11; Incom e < 2.5L; Merit based on previo us exams (YET cancel led).	75k - 1.5L p.a.	Jul-Au g	yet.nta .ac.in	⁵¹	2026- 08-03
schol_ nmms	Nation al Means	MoE	Class 8 pass;	needs Verific ation:	Aug-N ov	schola rships. gov.in	¹⁰	2026- 08-03

	-cum-Merit Scholarship		Govt school ; Parental income ceiling ; Minimum 55% marks.	true				
schol_nsp_postmatric	Post Matric Scholarships Scheme for Minorities	MoMA	Minority communities ; Class 11 to Ph.D; Income ceilings apply; Minimum 50% marks.	Variable	Aug-Oct	scholarships.gov.in	¹⁰	2026-08-03

Strategic Telemetry: Re-architecting the Student Journey

The efficacy of an educational guidance system is inextricably linked to the precise timing and psychological appropriateness of its data collection vectors. An analysis of the Indian educational continuum dictates a fundamental redesign of the application's telemetry. The objective is progressive profiling—requesting only the data required to facilitate the immediate next phase of decision-making, thereby reducing cognitive load and preventing user alienation. The initial onboarding phase is currently burdened with premature interrogations. Requesting detailed financial income brackets, nuanced caste category delineations, and disability statuses

from a Class 9 student upon first launching the application creates unnecessary friction. These data points do not fundamentally alter the broad career exploration phase; a student's initial exposure to engineering, healthcare, or civil service pathways should not be algorithmically gated by their financial status. Therefore, demographic and financial telemetry must be aggressively deferred. This data should only be requested contextually—specifically, at the point of action, such as when a user explicitly requests to view scholarship eligibility criteria or attempts to calculate examination fee relaxations.

The essential data requirements mutate radically across the developmental timeline:

1. **Class 9 to 10:** The telemetry must focus purely on board affiliation and broad, macro-level interests (as defined in the Two-Tier taxonomy). Understanding whether a student is under the WBBSE or CBSE directly informs the application of their upcoming structural realities, without requiring them to prematurely declare a professional allegiance.
2. **Class 11 to 12:** The cognitive demands shift sharply toward structural rigidities. At this juncture, asking for generalized interests is entirely insufficient. The application must interrogate the student regarding their specific stream selection (Science, Commerce, Humanities, Vocational) and, critically, their exact subject combination matrices. As evidenced by the WBCHSE framework, failing to ask a Class 11 student about their specific elective subset restricts the application to rendering highly generalized and potentially inaccurate post-secondary guidance³⁴. The application must map their selected subjects to downstream examination eligibility.
3. **Vocational and Diploma Routes:** For students navigating the vocational corridors, the questioning architecture must pivot toward lateral mobility and apprenticeship readiness. The application currently fails to inquire about a diploma student's intent to pursue lateral entry into baccalaureate programs. By asking about direct second-year entry into a B.Tech program, the system can dynamically surface state-specific examinations like the OJEE Lateral Entry⁶. Interrogating ITI students regarding their interest in the National Apprenticeship Promotion Scheme (NAPS) bridges the gap between theoretical vocational training and immediate industrial deployment¹⁴.
4. **Undergraduate and Post-Graduate:** Generalized questioning becomes obsolete. The application must ask about specific degree disciplines, graduation years, and precise professional aspirations. A student targeting academia must be identified early to surface UGC NET timelines⁵⁵, while a student aiming for immediate employment in the banking sector must be routed toward SBI PO timelines⁵⁶. The success of the application relies on acknowledging that a 21-year-old graduate requires fundamentally different architectural routing than a 15-year-old matriculant.

Telemetry Gaps and Unverified Domains

The strict mandate governing this dataset requires the explicit demarcation of unverified metrics. Filling gaps with plausible assumptions violates the core architectural rule. During the verification process, several explicit gaps were identified where official documentation either obfuscates the data or delays its publication:

- **Examination Fees:** Registration fees for major national examinations fluctuate annually and are often highly stratified by exact caste and income intersections. Therefore, static fee data for the JEE Main, NEET UG, and SBI PO exams have been flagged as needsVerification: true. The application should utilize dynamic pointers to official notifications rather than hardcoding fluid financial requirements.
- **Exact Category Relaxations:** While overarching relaxations are documented (e.g., UGC NET offering a 5% relaxation for reserved categories, or SBI PO offering specific age relaxations¹⁸), granular percentage threshold differences across all state-level polytechnic entrance exams remain undocumented in the provided research scope and must be flagged for manual verification by the user at runtime.
- **State-Specific Scholarship Stipends:** Exact disbursement amounts for the NMMS and minor state schemes require annual verification against the scholarships.gov.in portal, as state matching funds alter the final beneficiary amounts¹⁰.

By maintaining this rigid standard of provenance, the application secures its position as a trustless, reliable arbiter of career trajectories, ensuring no student is routed toward a discontinued exam, a closed admission window, or a fictional coaching product.

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